

# The 2014–2016 Oil Price Collapse as a Labor Demand Shock to Oklahoma

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August 5, 2026

## **Abstract**

Between June 2014 and February 2016, the spot price of West Texas Intermediate crude fell by 71%, from \$105.79 to \$30.32 per barrel, an externally-caused collapse in the value of marginal product of labor in oil-producing regions. This paper treats the collapse as a natural experiment in state labor market equilibrium by estimating effects in Oklahoma, one of the most oil-intensive state economies, using a difference-in-differences (DiD) design with neighboring, non-oil Arkansas as the comparative benchmark, as per Card and Krueger (1994). Using monthly BLS data retrieved from FRED for 2010-2019 (240 state-month observations), I find that Oklahoma's seasonally adjusted unemployment rate rose by 1.78% relative to Arkansas after January 2015 (Newey-West SE = 0.22). The effect peaks at 2.2% in 2016, coincides with a 14.6% contraction in Oklahoma's mining, logging, and construction employment, and survives alternative treatment dates, state-specific trends, pre-trend extrapolation, and a pre-period placebo test. The results match the textbook prediction that a negative labor demand shock lowers equilibrium employment, and they carry direct implications for how commodity-dependent states such as Oklahoma manage fiscal buffers and workforce transitions.

## **1 Introduction**

Labor economists cannot run controlled experiments on entire labor markets, so we look for moments when history runs one for us. The Mariel boatlift gave Card (1990) a sudden supply shock; New Jersey's 1992 minimum wage gave Card and Krueger (1994) a policy shock identified against neighboring Pennsylvania. This paper studies a demand shock of comparable size: the collapse of world oil prices that began in mid-2014, when OPEC declined to cut production amid surging U.S. shale supply and weakening global demand (Baumeister

and Kilian, 2016). West Texas Intermediate (WTI), priced at Cushing, Oklahoma, fell from \$105.79 per barrel in June 2014 to \$30.32 in February 2016, a 71% decline in 20 months.

In a state like Oklahoma, where oil and gas extraction is a large share of output and employment, halving the oil price halves the value of marginal product of labor in one of its largest industries. The research question is therefore direct: how did a large, plausibly exogenous negative labor demand shock affect equilibrium employment and unemployment in Oklahoma’s labor market?. I answer it with a DiD design comparing Oklahoma (OK) to Arkansas (AR), a neighboring state with a similar size, region, and labor market trajectory but no significant oil and gas extraction sector.

The headline result is that the oil bust raised Oklahoma’s unemployment rate by 1.8% compared to Arkansas over 2015–2019. The gap peaks at 2.2% in 2016 before partially closing as prices and drilling recovered. The mechanism is visible in the payroll data: employment in Oklahoma’s mining, logging, and construction fell 14.6% from its December 2014 peak to its August 2016 trough. A placebo treatment placed in the middle of the pre-period produces an effect statistically indistinguishable from zero.

The remainder of the report proceeds as follows: Section 2 lays out the theory, Section 3 describes the data, Section 4 presents empirical strategy, Section 5 reports results, Section 6 discusses Oklahoma policy implications, and Section 7 concludes.

## 2 Background & Theory

### 2.1 Oil Prices & Labor Demand

In the competitive model of Chapter 4, a state labor market clears where labor supply equals labor demand, and each point on the demand curve satisfies  $w = VMP_E = p \cdot MP_E$ : firms hire until the wage equals the value of the marginal product of labor (Borjas, 2016). The oil price enters this through  $p$ . When the crude price halves, the marginal barrel a drilling crew produces is worth half, so at any wage the profit-maximizing level of employment in the oil-exposed sector falls. Figure 1 shows the prediction, where the demand curve shifts inward from  $D_0$  to  $D_1$ , and equilibrium moves from  $(E_0, w_0)$  to  $(E_1, w_1)$ , lower employment/wages.

Three channels amplify the state-level effect beyond the direct sectoral one: First, oil extraction has long local supply chains (pipeline construction, equipment, trucking, hospitality for crews) so the demand shock propagates to non-tradable sectors through local spending; Feyrer et al. (2017) estimate that each dollar of new oil and gas production during the fracking boom raised total local income by more than the direct wage bill, and the bust runs that multiplier in reverse. Second, the adjustment through migration is slow. Migration

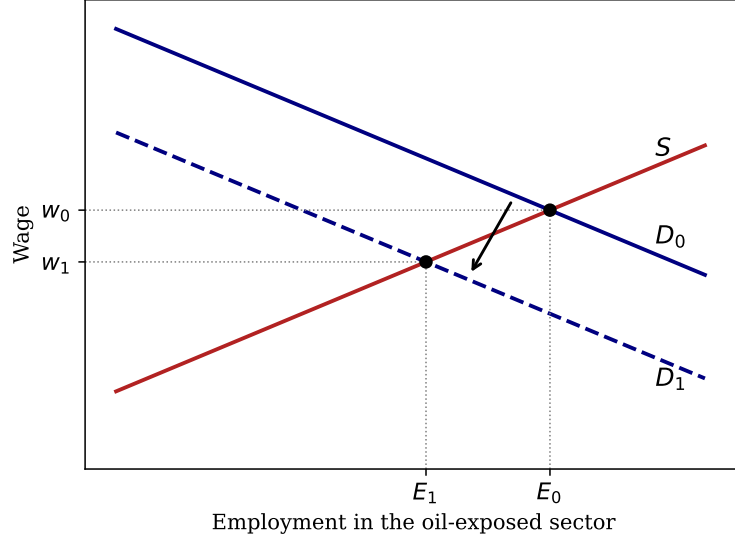


Figure 1: Negative product-price shock as the fall in the oil price lowers  $VMP_E = p \cdot MP_E$  at every employment level, shifting labor demand from  $D_0$  to  $D_1$ ; equilibrium employment falls from  $E_0$  to  $E_1$  and the wage from  $w_0$  to  $w_1$ .

costs are large, so in the short run the burden of a local demand shock shows up in local unemployment and wages rather than in immediate outflows (Blanchard and Katz, 1992). Third, energy busts reduce state government revenue from gross production taxes, forcing procyclical public sector cuts.

## 2.2 Why is this a credible natural experiment?

Evident in the 2014-2016 collapse is that no plausible channel runs from Oklahoma's labor market to the world oil price. The decline was driven by global supply and demand forces; the U.S. shale boom, OPEC's November 2014 decision not to cut output, and softening world activity (Baumeister and Kilian, 2016; Kilian, 2009). Oklahoma is a price taker. The shock is external to Oklahoma's labor market, sharply timed, and enormous — the same ingredients Card (1990) exploited in the Mariel episode, but on the demand side of the market.

The design follows Card and Krueger (1994): identify the effect of the shock by comparing the treated state's change in outcomes to the change in a neighboring comparison state that shares regional macro conditions but not the treatment. Arkansas borders Oklahoma, has a similar population and industry mix outside of energy, is subject to the same regional and national business cycle, but has essentially no oil and gas extraction sector. If national forces drive both states equally, the difference in their changes isolates the oil shock.

Table 1: Summary statistics, estimation window (Jan 2010-Dec 2019,  $N = 120$  months per series). Unemployment rates are seasonally-adjusted percentages of labor (BLS LAUS via FRED); employment is thousands; WTI is dollars per barrel (FRED, WTISPLC).

|                                        | Mean   | Std. dev. | Min    | Max    |
|----------------------------------------|--------|-----------|--------|--------|
| OK unemployment rate (%)               | 4.59   | 0.98      | 3.00   | 6.90   |
| AR unemployment rate (%)               | 5.57   | 1.75      | 3.40   | 8.00   |
| OK mining/logging/constr. empl. (000s) | 127.57 | 8.23      | 105.90 | 140.80 |
| WTI spot price (\$/bbl)                | 72.45  | 21.94     | 30.32  | 110.04 |

## 3 Data

### 3.1 Sources and sample

All series are monthly, seasonally adjusted, and retrieved from FRED (Federal Reserve Bank of St. Louis) in August 2026. The four series, with FRED identifiers, are the Oklahoma unemployment rate (OKUR, BLS Local Area Unemployment Statistics), the Arkansas unemployment rate (ARUR), all employees in mining, logging, and construction in Oklahoma (SMS40000001500000001, BLS Current Employment Statistics), containing oil and gas extraction and drilling/construction activity, and the WTI spot price (WTISPLC). A reader can rebuild the dataset exactly by downloading these four identifiers from [fred.stlouisfed.org](https://fred.stlouisfed.org) for 2005-2019; the accompanying code ships the extract as four '.csv' files.

The unit of observation in the estimation sample is the *state-month*. The sample runs from January 2010 to December 2019:  $120 \text{ months} \times 2 \text{ states} = 240$  observations, split into a 60-month pre-period (2010-2014) and a 60-month post-period (2015-2019). I start in 2010 to avoid conflating the oil bust with the Great Recession's collapse-and-recovery dynamics, and I end in December 2019 to prevent COVID-19 events from affecting the sample. The mining series and WTI are used descriptively from 2005 onward.

### 3.2 Descriptive Evidence

Table 1 reports summary statistics. Figure 2 shows the first stage shock: WTI collapses post-2014, and Oklahoma's mining, logging, and construction employment falls from its December 2014 peak of 140,800 to 120,200 by August 2016, a 14.6% contraction, before partially recovering and dipping again in 2019.

Figure 3 previews the main result. Through 2010-2014 the two states' unemployment rates fall roughly in parallel, with Oklahoma persistently below Arkansas. Beginning in early 2015 the paths cross: Oklahoma's unemployment rate rises from 4.0 to 4.8% during 2015-2016 while Arkansas' continues falling from 5.5 to 4.0%. The Oklahoma-minus-Arkansas

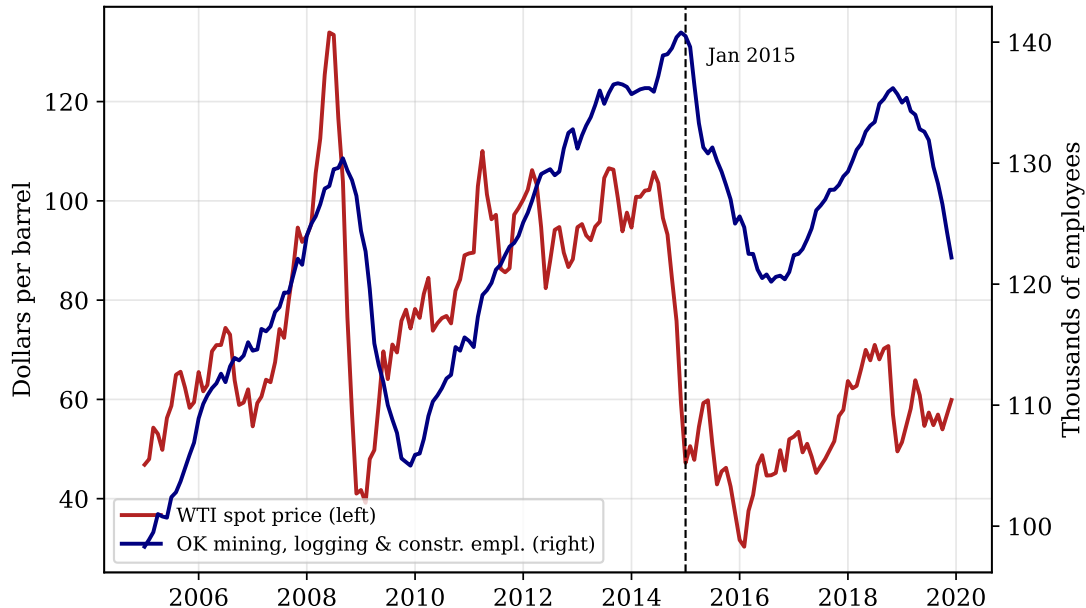


Figure 2: The shock and the first stage, 2005-2019. WTI spot price (left axis, FRED WTISPLC) and Oklahoma mining, logging, and construction employment (right axis, FRED SMS40000001500000001, seasonally adjusted). Dashed line marks the January 2015 treatment date.  $N = 180$  months.

gap,  $-1.87$  points on average pre-2015, closes to zero post-2015 (Figure 5). This suggests a large adverse relative shock to Oklahoma timed exactly to the oil bust.

## 4 Empirical Strategy

The estimating equation is the two-group, two-period DiD applied to a monthly panel:

$$UR_{st} = \alpha + \beta_1 OK_s + \beta_2 Post_t + \delta (OK_s \times Post_t) + \varepsilon_{st}, \quad (1)$$

where  $UR_{st}$  is the adjusted unemployment rate of state  $s \in \{OK, AR\}$  in month  $t$ ;  $OK_s$  is an indicator for Oklahoma;  $Post_t$  is an indicator for months from January 2015 onward (WTI had fallen below \$50 and drilling activity had begun to decrease).  $\delta$  is the parameter of interest, and I replace  $Post_t$  with a full set of month fixed effects  $\lambda_t$ , and estimate an event-based instance that replaces the interaction with  $OK_s \times$  year dummies, normalized to 2014, to trace and see pre-trends.

**Identifying assumption.**  $\delta$  has a causal interpretation under parallel trends: absent the oil collapse, Oklahoma's unemployment rate would have evolved like Arkansas'. This is plausible here since the two states track each other closely over 2010-2014 (Figure 3).

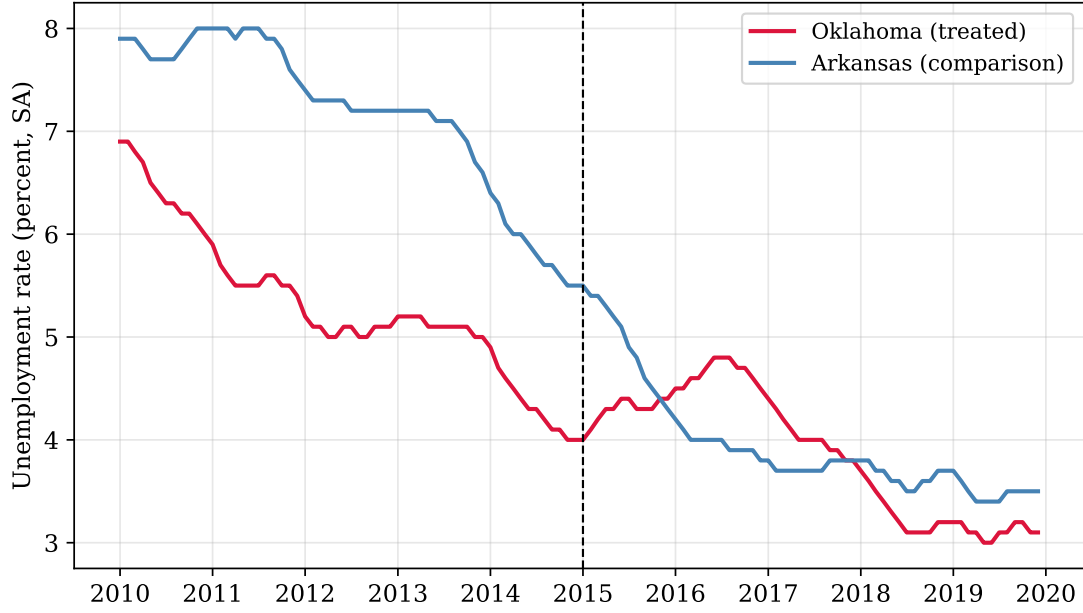


Figure 3: Seasonally adjusted unemployment rates, Oklahoma (treated) and Arkansas (comparison), Jan 2010-Dec 2019 (FRED OKUR, ARUR; BLS LAUS). Dashed line marks January 2015.  $N = 120$  months per state.

The shock's origin is global and could not have been caused by Oklahoma conditions. The main threats are, first, differential exposure to other shocks arriving around 2015 (I know of no Arkansas-specific boom or Oklahoma-specific non-oil bust at that date) and, second, spillovers that violate the stable unit treatment value assumption. If displaced Oklahoma workers migrated to Arkansas and raised labor supply, Arkansas' unemployment would be pushed up, biasing  $\hat{\delta}$  to zero. Conversely, if the bust in Oklahoma slowed Arkansas supplier demand, the estimate would bias away from zero. Arkansas' post-2015 trajectory, which matches the nation's, gives no sign of this.

**Inference.** With only these two states, clustering by state is impossible and serially correlated errors are the first-order inference problem in DiD settings (Bertrand et al., 2004). I therefore report Newey-West standard errors with a 12-month bandwidth, and as robustness I allow state-specific linear trends, shorten the window, move the treatment date, and run a placebo DiD entirely within the pre-period. The estimates should still be read as case study within Card-Krueger rather than an averaged causal parameter.

Table 2: Difference-in-differences in mean unemployment rates (%). Pre = Jan 2010-Dec 2014 (60 months per state); Post = Jan 2015-Dec 2019 (60 months per state). Data: FRED OKUR, ARUR.

|                           | Pre (2010–2014) | Post (2015–2019) | Change       |
|---------------------------|-----------------|------------------|--------------|
| Oklahoma (treated)        | 5.31            | 3.88             | -1.43        |
| Arkansas (comparison)     | 7.18            | 3.96             | -3.22        |
| Difference-in-differences |                 |                  | <b>+1.78</b> |

## 5 Results

### 5.1 Main estimates

Table 2 shows the transparent  $2 \times 2$  version of the design using pre/post means. Oklahoma’s unemployment rate fell 1.43% between the pre- and post-periods, and Arkansas’ fell by 3.22%. The DiD is +1.78 points: relative to its comparison, Oklahoma’s unemployment rate deteriorated by almost 2% after the oil collapse.

Table 3 reports regression estimates of equation (1). Column 1 is the baseline:  $\hat{\delta} = 1.783$  (Newey-West SE 0.216), significant at the 1% level. Column 2 replaces the post dummy with 120 month fixed effects since the panel has two groups, the point estimate is identical (1.783, SE 0.210), but the specification makes explicit that all common time shocks are absorbed. Column 3 adds state-specific linear trends, a demanding check given that trends absorb slow-moving relative movements: the estimate is essentially unchanged at 1.851 (SE 0.520). In economic terms, a 71% oil price collapse raised the unemployment rate of a heavily exposed state by 1.8% on a pre-period Oklahoma mean of 5.3%, about a one-third increase in the unemployment rate, and roughly 33,000 additional unemployed workers at Oklahoma’s labor force of about 1.8 million total.

### 5.2 Dynamics & Pre-trends

Figure 4 plots the event-study coefficients on  $OK \times \text{year}$ , normalized to 2014, with 95% confidence bands.

Three features stand out. First, the pre-period coefficients are small relative to the treatment effects but not uniformly zero: the OK-AR gap widened in 2011-2012 (by up to 0.78 points) as Oklahoma’s shale-boom recovery outpaced Arkansas’, then narrowed back by 2014, as the 2011-2014 narrowing runs in the same direction as the estimated effect. I treat it as the main identification concern and address it directly below; note already that the movement is a hump, not a secular drift, which is a linear trend fit to the full 2010-2014 gap series has a slope of only +0.01 points annually. Second, the effect appears immediately in

Table 3: Regression difference-in-differences estimates. Dependent variable: state unemployment rate (pp). Newey-West standard errors in parentheses in all columns; \*, \*\*, \*\*\* denote significance at 10, 5, and 1%. Columns 1-3: Jan 2010-Dec 2019, treatment date Jan 2015. Column 4 moves the treatment date to Jul 2014. Column 5 ends the sample in Dec 2016 (bust window only). Column 6 is a placebo estimated on the pre-period only (2010-2014) with a fake treatment date of Jul 2012.

|                  | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)              |
|------------------|---------------------|---------------------|---------------------|---------------------|---------------------|------------------|
|                  | Baseline            | Month FE            | State trends        | Post = Jul 14       | 2010–16             | Placebo          |
| OK $\times$ Post | 1.783***<br>(0.216) | 1.783***<br>(0.210) | 1.851***<br>(0.520) | 1.693***<br>(0.244) | 1.874***<br>(0.380) | 0.120<br>(0.234) |
| Time effects     | Post dummy          | Month FE            | Post + trends       | Post dummy          | Post dummy          | Post dummy       |
| Observations     | 240                 | 240                 | 240                 | 240                 | 168                 | 120              |
| $R^2$            | 0.804               | 0.974               | 0.959               | 0.822               | 0.744               | 0.827            |

2015 and peaks in 2016 at +2.21 points (Newey-West SE 0.08, though with one treated state such precision should be read cautiously) before decaying as WTI recovers into the \$50-70 range. Third, the effect does not return to zero by 2019, consistent with the renewed price slide and rig-count decline that year and with the slow-migration adjustment of Blanchard and Katz (1992).

### 5.3 Robustness

Columns 4-6 of Table 3 probe the result. Dating the treatment from July 2014, when prices first began to fall, yields 1.693 (SE 0.244). Restricting the sample to the bust window, so that the post-period recovery cannot dilute the estimate, yields 1.874 (SE 0.380). The placebo in column 6 returns 0.120 (SE 0.234,  $p = 0.61$ ): the design finds nothing where nothing should be found.

I address the 2011-2014 pre-period movement in two further ways. Column 3 allows each state its own linear trend estimated jointly with the treatment effect, leaving the estimate at 1.851 (SE 0.520). More conservatively, I fit a linear trend to the OK-AR gap using pre-period data, extrapolate it through 2015-2019 as the no-bust counterfactual, and measure the mean deviation of the actual post-period gap from that counterfactual. Since the fitted pre-period drift is essentially flat (+0.01 points annually), this trend-adjusted estimate is 1.75 points, which is indistinguishable from the baseline. The parallel-trends assumption is therefore not exactly satisfied year by year, but no reasonable extrapolation of the pre-period relationship comes close to generating the post-2014 divergence.

Two caveats temper the causal reading. The comparison involves a single treated and a single comparison state, so idiosyncratic Arkansas shocks cannot be averaged away; and

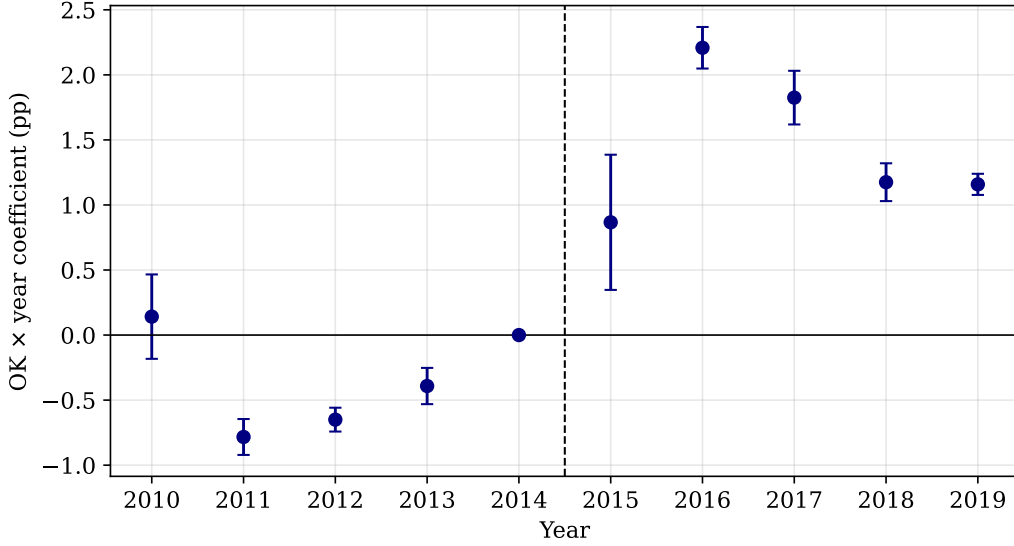


Figure 4: Event-study estimates: coefficients on  $OK \times \text{year}$  from equation (1) with year effects, base year 2014. Bars are 95% confidence intervals using Newey-West (12-lag) standard errors.  $N = 240$  state-months.

because Arkansas' post-2015 unemployment decline mirrors the national decline, part of the estimated  $\delta$  could reflect Oklahoma merely failing to share a national improvement for non-oil reasons. The tight coincidence of timing between WTI, Oklahoma drilling employment, and the unemployment gap makes a non-oil explanation strained, but the estimate is best summarized as the causal effect of the oil bust on Oklahoma relative to a plausible no-bust counterfactual, measured with the precision a two-state design allows.

## 6 Policy Discussion

The findings speak directly to Oklahoma. First, they quantify the state's macro-labor exposure to a single commodity price: one oil cycle moved the state unemployment rate by nearly two points relative to trend, an order of magnitude larger than typical estimates of state policy effects. For a state that funds core services partly from gross production taxes, the result strengthens the case for counter-cyclical fiscal buffers, Oklahoma's Rainy Day Fund and the 2016-2018 budget crises are the institutional embodiment of this lesson, and for revenue diversification, since the same shock that destroys private paychecks also destroys the tax base needed to cushion it.

Second, the mechanism points at workforce policy. The employment losses were concentrated in drilling-linked work: mining, logging, and construction payrolls fell 20,000 jobs in

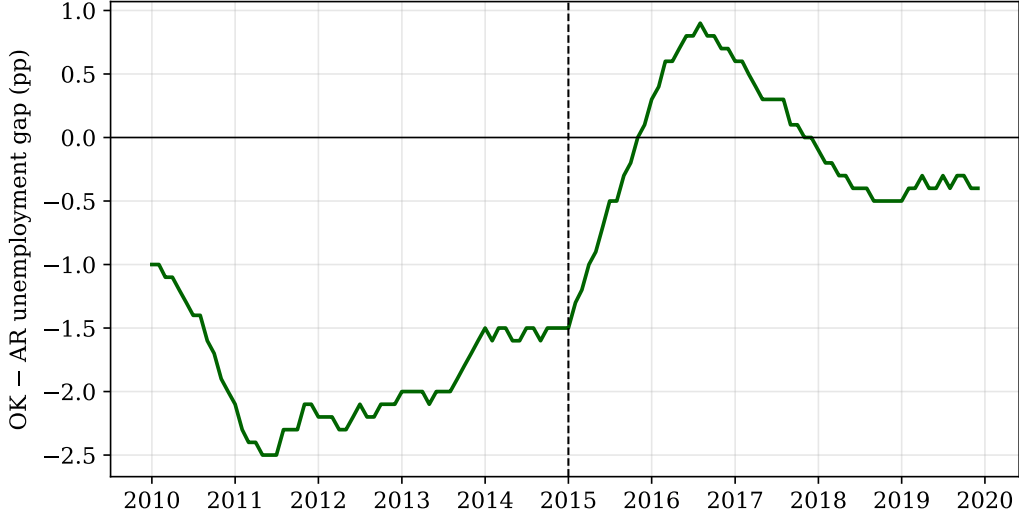


Figure 5: The Oklahoma-minus-Arkansas unemployment gap, Jan 2010-Dec 2019. The pre-period mean gap is  $-1.87$  points; the post-period mean is  $-0.09$  pp.  $N = 120$  months. Data: FRED OKUR, ARUR.

twenty months. Evidence that unemployment elevated for years rather than months suggests the migration margin worked slowly, as Blanchard and Katz (1992) found for U.S. states generally. That raises the value of retraining and reemployment services that move displaced rig and completion workers into sectors with transferable skill demand, and of unemployment insurance rules calibrated to multi-year energy cycles rather than ordinary recessions.

Third, energy firms can and do hedge price risk, but workers cannot. To the extent that hedging, diversification into midstream activity, and flexible-cost drilling (shorter-cycle shale investment) mute the employment response of future busts, the estimated elasticity here is an upper bound for the next cycle, but the 2019 dip in Oklahoma drilling employment at merely \$55 oil suggests the exposure remains live. What this study does not establish is the wage side of the adjustment or worker-level incidence. State-level rates cannot reveal who bore the shock, for how long, or at what earnings cost.

## 7 Conclusion

This paper asked how the 2014-2016 oil price collapse—a 71%, globally driven decline in the price of Oklahoma’s flagship product—affected equilibrium in Oklahoma’s labor market. Using monthly BLS data from FRED and a Card-Krueger-style difference-in-differences design against neighboring Arkansas, I estimate that the bust raised Oklahoma’s unemployment rate by 1.8%, peaking at 2.2% in 2016, alongside a 14.6% contraction in drilling-linked

employment. The estimate is robust to alternative treatment dates, state trends, and a pre-period placebo. The natural next step is to bring worker-level data (CPS or LEHD) to the same design to measure the earnings losses, industry transitions, and migration responses of the displaced workers themselves.

## References

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## Appendix: Reproducibility

All numbers, tables, and figures in this report are produced by `code/make_data.py` (which writes the FRED extract to `data/*.csv`) included in running the notebook `analysis.ipynb` top to bottom. Relative paths are used throughout. Data provenance: FRED series OKUR, ARUR, SMS40000001500000001, and WTISPLC, retrieved August 2026.